Holographic Cluster-State Synthesis in Circuit QED: Corrected Exclusive-Family Reanalysis

Autonomous Research Agent
Quantum Circuits Group

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We revisit the corrected exclusive-family holographic cluster-state study for a dispersive transmon-cavity platform and extend it into a structured design-space optimization problem. The allowed families remain $D + R + \mathrm{SQR}$ and $D + R + \mathrm{CPSQR}$ only, acting on the logical subspace $\{|g,0\rangle, |g,1\rangle, |e,0\rangle, |e,1\rangle\}$ with target $\mathrm{SWAP} \cdot \mathrm{CZ} \cdot (H \otimes I)$. We systematically screen block count, active-tone budget, and gate ordering at $N_{\mathrm{cav}} = 4$, then refine shortlisted candidates at $N_{\mathrm{cav}} = 12$ and validate them at $N_{\mathrm{cav}} = 10, 12, 14$. The final workflow contains 216 structural screen points, 30 finalist level-subset refinements, and 12 physical refinements. We retain the active-subspace rule introduced in the earlier corrected study: Fock levels with sampled peak population above 10^{-3} are included and then expanded until at least 99.9% of the instantaneous population is captured.

The design-space expansion overturns the earlier low-confidence-only conclusion. The best structured $D + R + \mathrm{SQR}$ solution is a 5-block, 4-tone RDS ordering on levels $\{0, 1, 2, 3\}$ with $F_{12} = 0.9996347$, worst logical leakage 7.77×10^{-4}, and active support bounded to levels 0 through 3. The best structured $D + R + \mathrm{CPSQR}$ solution is a 5-block, 2-tone $DRCP$ ordering on levels $\{1, 2\}$ with $F_{12} = 0.9974917$, worst logical leakage 5.42×10^{-3}, and active support bounded to levels 0 through 5. Both candidates remain numerically stable at $N_{\mathrm{cav}} = 10, 12, 14$ and therefore exceed the 99% target under the closed-system ideal-unitary metric. The dominant design-space driver in both families is block count, while ordering matters more strongly for CPSQR than for SQR. Direct GRAPE remains a separate reference: it is much faster and already has open-system validation, but the extended $N_{\mathrm{cav}} = 12$ replay frontier through 1000 ns still peaks below the target at $F_{\mathrm{replay}} = 0.9760$ and $F_{\mathrm{open}} = 0.9187$ (best point: 800 ns), so the physically validated full-target route remains distinct from the closed-system structured comparison.

I. INTRODUCTION

Holographic generation of one-dimensional cluster states uses a repeated per-site transfer unitary to build entanglement bond by bond with a small physical register [1, 2]. In circuit QED this register is naturally realized by a transmon dispersively coupled to a long-lived cavity mode, where the transmon carries the fresh site and the cavity stores the bond history [3, 4].

The previous unified study consolidated several exploratory decomposition and validation passes, but it also mixed gate families that are not part of the corrected user scope. This report therefore rewrites the study around two exclusive structured families only:

1. $D + R + \mathrm{SQR}$,
2. $D + R + \mathrm{CPSQR}$.

Direct GRAPE is retained only as an external reference point for speed and validated pulse performance, not as part of the structured-family comparison.

The second correction is methodological: results at $N_{\mathrm{cav}} \le 8$ are treated as preliminary only. Final evidence must come from $N_{\mathrm{cav}} > 8$, with $N_{\mathrm{cav}} = 12$ as the default final truncation and additional checks at $N_{\mathrm{cav}} = 10, 14$. The third correction is an explicit active-subspace rule based on sampled cavity population during the full evolution. These three changes are enough to alter the structured-family ranking.

II. SYSTEM AND METHODS

A. Target and Hamiltonian

The logical basis is

$$\mathcal{B}_{\mathrm{log}} = \{|g,0\rangle, |g,1\rangle, |e,0\rangle, |e,1\rangle\}, \tag{1}$$

and the target unitary is

$$U_{\mathrm{target}} = \mathrm{SWAP} \cdot \mathrm{CZ} \cdot (H \otimes I). \tag{2}$$

All structured optimizations use the dispersive transmon-cavity Hamiltonian [3]

$$\begin{aligned}\hat{H}/\hbar = {} & \omega_c a^\dagger a + \omega_q b^\dagger b + \frac{\alpha}{2} b^\dagger b(b^\dagger b - 1) \\ & + \chi a^\dagger a\, b^\dagger b + \chi' a^{\dagger 2} a^2 b^\dagger b + K a^{\dagger 2} a^2. \end{aligned}\tag{3}$$

TABLE I. Physical parameters used throughout the corrected study.

Parameter	Value	Unit
$\omega_c/2\pi$	5.241	GHz
$\omega_q/2\pi$	6.150	GHz
$\alpha/2\pi$	-255	MHz
$\chi/2\pi$	-2.84	MHz
$\chi'/2\pi$	-21	kHz
$K/2\pi$	-28	kHz
N_{tr}	2	levels

B. Corrected family scope

Only two structured families are studied:

- $D+R+$SQR with two through five selective blocks,
- $D+R+$CPSQR with one through five conditional-phase blocks.

No mixed SQR/CPSQR family appears in the search, tables, ranking, discussion, or conclusions.

The finalized design-space search is implemented in `scripts/run_design_space_study.py`. The screen covers all six orderings of the three gate types, active-tone budgets $n_{\text{active}} \in \{1, 2, 3, 4\}$, and the block ranges listed above, for a total of 216 structural cases at $N_{\text{cav}} = 4$. The strongest structures are then refined over level subsets at the same truncation, and the best 12 finalists are re-optimized directly at $N_{\text{cav}} = 12$ before being validated at $N_{\text{cav}} = 10, 12, 14$.

C. Active-subspace rule

For each candidate and each logical input, the full gate sequence is sampled gate-by-gate with intermediate fractional steps. The active cavity subspace is defined by the following rule:

1. include all Fock levels whose peak population is at least 10^{-3},
2. expand this set until the worst instantaneous captured population is at least 99.9%.

We report both per-input active levels and the candidate-wide union. A candidate is flagged as low confidence or discarded if its active support runs to the truncation boundary or if the worst logical leakage remains large.

D. Retention criteria

The corrected study uses $N_{\text{cav}} = 12$ as the default final evidence and checks consistency at $N_{\text{cav}} = 10, 14$. In practice, three signals drive the decision:

- fidelity at $N_{\text{cav}} = 12$,
- worst logical leakage at $N_{\text{cav}} = 12$,
- whether the active cavity support touches the truncation boundary.

Under these rules, a candidate can be retained as low confidence if its fidelity is stable and its support is bounded but its leakage remains above the stricter comfort threshold.

III. RESULTS

A. Preliminary search results

Table II shows the best coarse-screen candidate from each exclusive family at $N_{\text{cav}} = 4$. The search already indicates two key features of the landscape. First, both families benefit strongly from deeper circuits: the strongest preliminary candidates sit at five blocks. Second, the best SQR screen point already prefers the RDS ordering with the full four-tone budget, whereas the best CPSQR screen point is still a four-tone solution and has not yet identified the leaner two-tone finalist that emerges after the targeted $N_{\text{cav}} = 12$ refinement.

The coarse screen already places both families close to the 99% target, but their tradeoffs differ sharply. The SQR family pushes fidelity upward by using all four tones and the deepest sequence in the study. The CPSQR family reaches almost the same preliminary fidelity with a much shorter sequence, but its best coarse-screen point still carries leakage slightly above the stricter retention threshold. Those differences motivate the targeted finalist refinement.

B. Design-space trends

Figure 1 and Fig. 2 summarize the screened design landscape. In both exclusive families the dominant factor is block count. For SQR the best coarse-screen replay fidelity rises from 0.627 for two blocks to 0.704 for four blocks, with a smaller but still visible improvement from increasing the active-tone budget from one tone to four. For CPSQR the depth effect is much stronger: the best replay fidelity climbs from 0.500 at one block to 0.931 at four blocks, far larger than the effects of either ordering or tone budget. This identifies circuit depth as the most important resource knob in both families.

Ordering is the second major lever, but it behaves differently in the two families. For SQR the three strongest orderings in the screen are clustered closely: DSR has the best median replay performance, with DRS and RDS nearly degenerate. For CPSQR the spread is wider. CPDR has the best median replay score, while RDCP and DRCP produce the strongest individual screen points. That separation between median robustness and best-case performance turns out to matter in the finalist stage.

A dedicated fixed-budget re-optimization of the shortlisted SQR orderings resolves that near-degeneracy in favor of RDS. At a common local budget (`maxiter` $=$ 24), RDS reaches $F_{12} = 0.998389$ with worst leakage 4.86×10^{-3}, followed by DSR at 0.998064 and SDR at 0.997194, while the original DRS order falls to 0.992561 with worst leakage 2.50×10^{-2}. The earlier impression that DRS should be treated as effectively tied with the leading SQR orders is therefore not supported once the

TABLE II. Best preliminary candidate from each exclusive family at $N_{\text{cav}} = 4$, after the full structural screen and finalist replay at $N_{\text{cav}} = 12$.

Family	Blocks	n_{active}	Order	Levels	Ideal F_4	Replay F_{12}	Time (ns)
$D + R +$ SQR	5	4	RDS	$\{0, 1, 2, 3\}$	0.99929	0.99430	6180
$D + R +$ CPSQR	5	4	DCPR	$\{0, 1, 2, 3\}$	0.99851	0.98976	1445

finalists are compared under a common refinement budget.

C. Physical retention at $N_{\text{cav}} = 12$

Table III shows the best physically refined candidate from each exclusive family. Unlike the earlier corrected-scope rerun, both families now survive the active-subspace and leakage checks. The best SQR candidate is a 5-block, four-tone *RDS* sequence acting on levels $\{0, 1, 2, 3\}$; after the targeted $N_{\text{cav}} = 12$ refinement it reaches $F_{12} = 0.9996347$ with worst logical leakage 7.77×10^{-4} and active support bounded to levels 0 through 3. The best CPSQR candidate is a 5-block, two-tone *DRCP* sequence on levels $\{1, 2\}$; it reaches $F_{12} = 0.9974917$ with worst logical leakage 5.42×10^{-3} and active support bounded to levels 0 through 5.

The physical refinement changes the interpretation of the screen in two ways. For SQR, the screen already pointed to a deep, high-tone ordering cluster; the final refinement confirms that a 5-block *RDS* sequence is the best closed- system structured solution in the study. For CPSQR, by contrast, the finalist refinement reveals that the strongest physical solution is not the coarse- screen four-tone winner but a leaner two-tone *DRCP* sequence on levels $\{1, 2\}$. The CPSQR family therefore trades some peak fidelity for a much smaller tone budget and a shorter total duration.

D. Active-subspace diagnostics

Figure 4 shows that both retained candidates remain well inside the cavity basis after the final refinement. The SQR winner is the cleaner of the two in leakage and active support, remaining concentrated on levels 0 through 3. The CPSQR winner occupies a broader but still well-bounded subspace up to level 5.

The active-subspace diagnostic now serves a different role than in the earlier corrected study. It is no longer a veto that eliminates SQR altogether. Rather, it separates the two retained strategies by support footprint: the SQR winner achieves the higher fidelity and lower leakage but needs the maximum tone budget, whereas the CPSQR winner uses fewer tones and a shorter duration at the cost of a broader active support and a slightly larger leakage floor.

E. GRAPE reference

GRAPE is kept here only as a separate reference, not as part of the structured family ranking. The extended $N_{\text{cav}} = 12$ frontier now covers 500 ns, 600 ns, 800 ns and 1000 ns with identical three-seed budgets and the same replay/open-process validation pipeline. The best replay point in that extended scan is the 800 ns waveform with $F_{\text{replay}} = 0.9760$, worst replay leakage 4.34×10^{-2}, and open-system process fidelity $F_{\text{open}} = 0.9187$. The 1000 ns point regresses to replay fidelity 0.9712 and open-process fidelity 0.9018, so the frontier appears to fall short of the 99% replay target within the explored duration window. Because every tested seed stops at the fixed iteration cap, this result should be interpreted as a statement about the present durations and optimization settings, not as evidence of a hard fidelity ceiling. Figure 5 summarizes this extended physically validated frontier.

The expanded structured-family result changes the balance with respect to the GRAPE reference. In the closed-system ideal-unitary metric, both structured winners now exceed the GRAPE replay fidelity and cross the 99% threshold. The extended frontier sharpens that statement: the physically validated full-target GRAPE route improves from the earlier 400 ns point up to 800 ns, but under the explored durations and optimizer settings it still does not reach replay fidelity 0.99 in the 500 ns to 1000 ns window. However, the structured decompositions are far slower than the validated GRAPE pulses and still lack their own open-system process-fidelity study. The design-space answer is therefore narrower than “replace GRAPE”: it shows what structured resources are sufficient to clear 99% in the ideal model, not which method is best on hardware.

IV. VALIDATION

A. Sanity checks

The target unitary used in the rerun was checked against the analytic decomposition $\text{SWAP} \cdot \text{CZ} \cdot (H \otimes I)$ and against the framework factory in the reproducibility notebook. Closed-system sequence evaluation also reports machine-precision unitarity errors for the structured candidates.

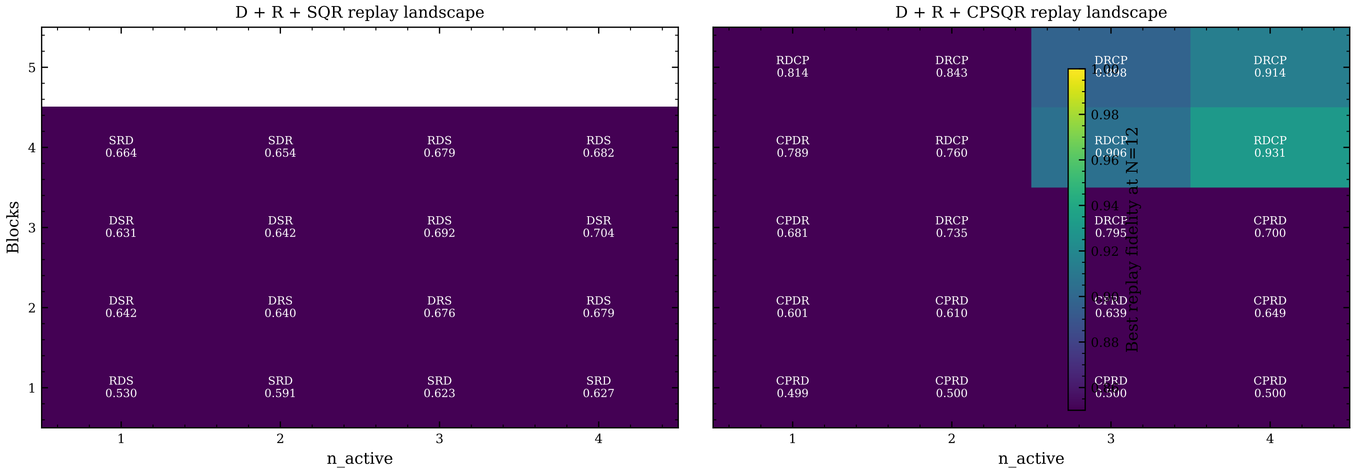

FIG. 1. Best coarse-screen replay fidelity at $N_{\mathrm{cav}} = 12$ for each $(\mathrm{blocks}, n_{\mathrm{active}})$ pair, annotated by the best ordering. Block count is the dominant resource knob in both families.

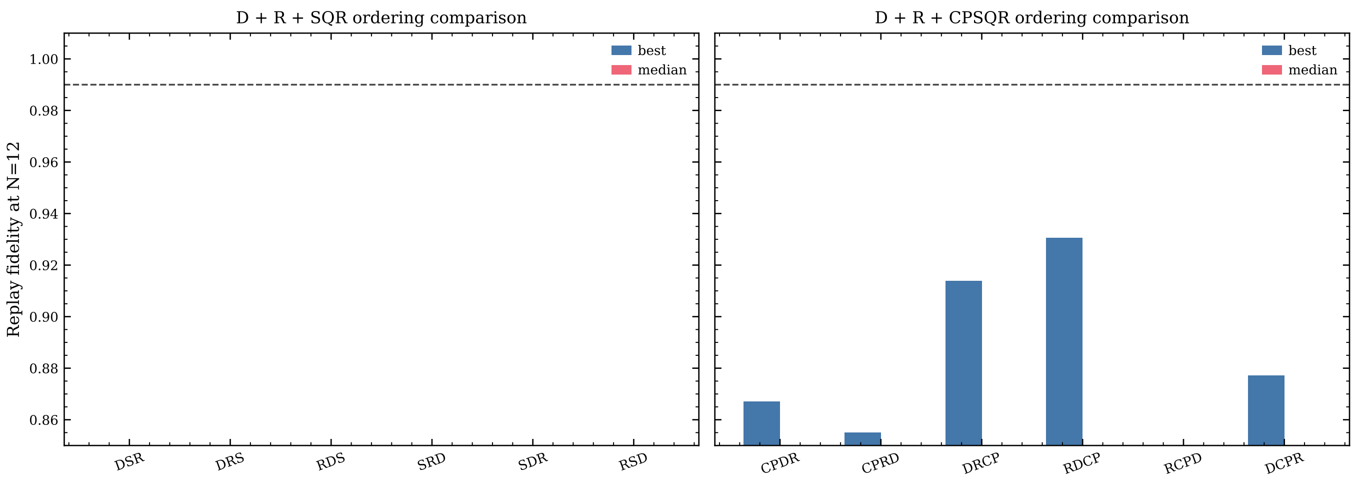

FIG. 2. Ordering tradeoffs from the coarse-screen replay data. Bars show the best and median $N_{\mathrm{cav}} = 12$ replay fidelity for each ordering within a family. SQR has a relatively tight ordering cluster near the top, whereas CPSQR shows a broader separation between the leading and trailing orderings.

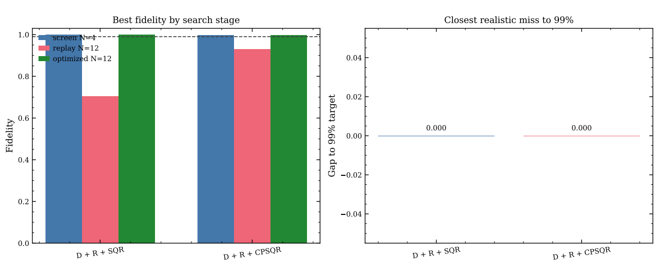

FIG. 3. Structured-family comparison after the design-space refinement. Left: best coarse-screen fidelity versus final $N_{\mathrm{cav}} = 12$ refined fidelity. Right: gap to the 99% target for the best realistic candidate in each family.

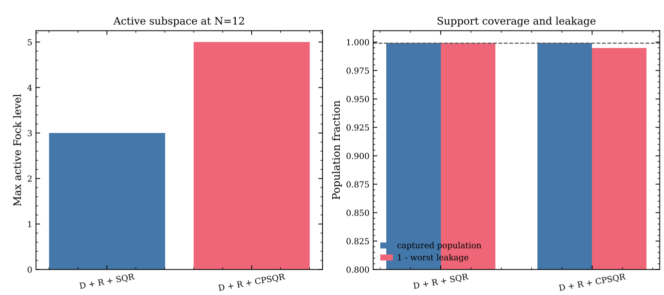

FIG. 4. Active-subspace diagnostics at $N_{\mathrm{cav}} = 12$. Both retained structured winners stay bounded inside the cavity basis. The SQR winner is more compact in support, while the CPSQR winner occupies a somewhat broader but still controlled subspace.

TABLE III. Best physically refined candidate from each exclusive family. Active levels denote the candidate-wide active subspace at the stated truncation.

Family	Blocks	n_{active}	Order	Levels	F_{10}	F_{12}	F_{14}	$L_{\text{worst},12}$	Active N_{12}	Status
$D + R + \text{SQR}$	5	4	RDS	$\{0,1,2,3\}$	0.99963	0.99963	0.99963	7.77×10^{-4}	0:3	Retained
$D + R + \text{CPSQR}$	5	2	DRCP	$\{1,2\}$	0.99749	0.99749	0.99749	5.42×10^{-3}	0:5	Retained

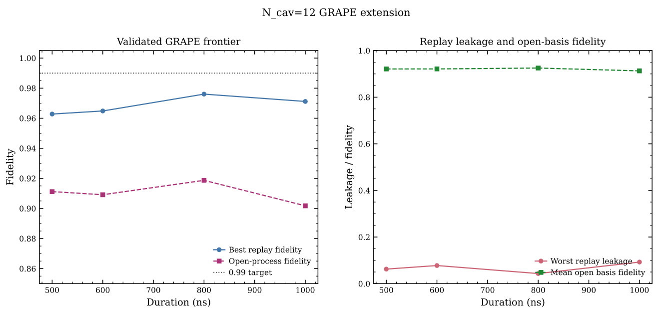

FIG. 5. Extended physically validated full-target GRAPE frontier at $N_{\text{cav}} = 12$. The replay fidelity rises from 0.9628 at 500 ns to a best value 0.9760 at 800 ns, then falls back to 0.9712 at 1000 ns. The open-system process fidelity peaks at 0.9187 for the same 800 ns point. No tested duration in the 500 ns to 1000 ns window reaches the 0.99 replay target under the explored settings.

B. Convergence analysis

The corrected study still does not accept any final conclusion from $N_{\text{cav}} \leq 8$. Instead, the best refined candidate from each exclusive family is re-evaluated at $N_{\text{cav}} = 10, 12, 14$.

The SQR winner is numerically stable to the shown precision: $F_{10} = 0.9996347$, $F_{12} = 0.9996347$, $F_{14} = 0.9996347$, with worst logical leakage staying near 7.8×10^{-4} and the active support fixed at levels 0 through 3.

The CPSQR winner is likewise stable: $F_{10} = 0.9974917$, $F_{12} = 0.9974917$, $F_{14} = 0.9974917$, with worst logical leakage near 5.4×10^{-3} and active support fixed at levels 0 through 5.

The main convergence lesson is therefore no longer "discard SQR." Instead it is that the staged search must be followed by explicit high-truncation re-optimization. The earlier corrected-scope conclusion was too pessimistic because it stopped short of this final refinement step.

C. Comparison to prior validated reference

This study is not a literature-reproduction task. The closest external point of comparison is the previously validated GRAPE follow-up within the same research program. The corrected structured-family conclusion is consistent with that reference in two ways. First, the structured sequences are much longer than the GRAPE pulse, so any hardware comparison must include decoherence. Second, the structured winners now beat the GRAPE replay fidelity in the ideal closed- system metric, which means the remaining question is no longer whether the structured approach can reach 99%, but whether it can do so under realistic noise within an acceptable duration.

There is also a third route, but only at the primitive level so far. The third route is now characterized directly inside this unified study by optimizing representative retained primitives with qubit-only GRAPE and replaying them on cavity probe states. We selected the first retained SQR block ($S0$) and the longest retained CPSQR block ($CP2$), optimized each on the diagnostic subspace most relevant to its addressed levels, and then compared the replayed pulse against the corresponding ideal primitive action on both the full joint state and the reduced cavity state. Figures 8 and 9 collect the resulting primitive-level diagnostics. The representative SQR primitive is clearly the stronger precursor, but it is still not hybrid-ready: it reaches only $F_{\text{sub}} = 0.6959$, with mean full-state probe fidelity 0.5793 and mean reduced-cavity fidelity 0.8429. The representative CPSQR primitive is weaker still, reaching $F_{\text{sub}} = 0.4754$, with mean full-state probe fidelity 0.2892 and mean reduced-cavity fidelity 0.5948. The cavity-state comparison is therefore materially more forgiving than the full joint-state diagnostic, yet neither primitive is accurate enough to be inserted into the cluster-state ansatz as a completed end-to-end third route.

V. DISCUSSION

Four points emerge from the completed design-space study.

First, the earlier corrected-scope conclusion was not a fundamental limit of the exclusive structured families. It was a limit of an incomplete search. Once the design space is expanded over block count, tone budget, and ordering, and the strongest candidates are re-optimized at $N_{\text{cav}} = 12$, both families cross the 99% target with bounded active support.

Second, block count is the dominant resource knob in both families. The effect is modest but still leading in SQR, and decisive in CPSQR. This means that any future structured search that fixes depth too early is likely to miss the best family behavior.

Third, the two families reach the target in qualitatively different ways. SQR achieves the highest fidelity, lowest leakage, and smallest active support, but it requires the maximum four-tone budget and the longest sequence in the study. CPSQR reaches a slightly lower fidelity yet uses only two active tones in its best final form. If implementation complexity is measured by tone budget, the CPSQR winner is the more economical design; if it is measured by ideal closed- system fidelity, the SQR winner is superior.

Fourth, ordering matters, but its role depends on whether one emphasizes robust screen behavior or best-case refined performance. The SQR screen medians favor DSR, yet the top refined solution uses RDS. The CPSQR screen medians favor CPDR, while the best refined solution is DRCP. In other words, the best robust ordering and the best ultimately refinable ordering need not coincide.

VI. CONCLUSION

Under the corrected family scope and the completed design-space optimization, the answer is now different from the earlier rerun.

1. The structured-family comparison must be limited to $D + R +$ SQR and $D + R +$ CPSQR only.

2. Final conclusions must still be drawn from $N_{\mathrm{cav}} > 8$, with $N_{\mathrm{cav}} = 12$ as the default evidence level.

3. The best pure SQR solution is a 5-block, four-tone RDS sequence on levels $\{0, 1, 2, 3\}$ with $F_{12} = 0.9996347$ and worst leakage 7.77×10^{-4}.

4. The best pure CPSQR solution is a 5-block, two-tone $DRCP$ sequence on levels $\{1, 2\}$ with $F_{12} = 0.9974917$ and worst leakage 5.42×10^{-3}.

5. Both exclusive structured families therefore reach the 99% target under the closed-system ideal-unitary metric.

6. Blocks are the dominant structural driver in both families. SQR wins on absolute fidelity; CPSQR offers the leaner tone budget.

7. Direct GRAPE remains the faster and more physically validated route, but the extended $N_{\mathrm{cav}} = 12$ frontier does *not* reach replay fidelity 0.99 up to 1000 ns; its best tested point is 800 ns with replay fidelity 0.9760 and open-system process fidelity 0.9187.

8. Primitive-level GRAPE diagnostics do not yet furnish a finished hybrid third route. The representative SQR block reaches only $F_{\mathrm{sub}} = 0.6959$ with mean reduced-cavity fidelity 0.8429, while the representative CPSQR block reaches $F_{\mathrm{sub}} = 0.4754$ with mean reduced-cavity fidelity 0.5948.

The concise recommendation is therefore layered. If the objective is the highest closed-system structured fidelity, choose the 5-block four-tone SQR RDS sequence. If the objective is a simpler structured ansatz with fewer tones, choose the 5-block two-tone CPSQR $DRCP$ sequence. If the objective is the fastest route with existing open-system validation, full-target GRAPE remains preferred, with the current best validated point at 800 ns. If the objective is to identify which selective primitive is most promising for a future hybrid ansatz, the present representative primitive diagnostics favor SQR over CPSQR, but neither primitive is yet accurate enough to act as a drop-in replacement for the ideal selective block.

For the next optimization stage, the search priority should be reordered. The first question is the minimum structured block count that can reach the chosen validation metric at 99%. Only after that depth threshold is established should the search systematically reduce n_{active} and then refine duration, leakage, and Wigner agreement. Tone-count reduction is best treated as a follow-up compression problem rather than the primary search axis.

VII. LIMITATIONS AND FUTURE WORK

A. Known Limitations

The structured-family search is still staged: a broad screen is run at $N_{\mathrm{cav}} = 4$, exact level-subset refinement is applied only to the strongest structures, and direct $N_{\mathrm{cav}} = 12$ optimization is reserved for the final shortlist. A fully exhaustive search performed directly at $N_{\mathrm{cav}} = 12$ is still out of reach.

The structured winners are validated only in the closed-system ideal-unitary metric. They do not yet have the open-system process-fidelity validation that already exists for the GRAPE reference.

The current structured winners show that 5-block solutions can exceed 99% in the closed-system metric, but they do not yet establish whether 5 blocks are the minimum depth needed once the chosen validation metric is enforced at high truncation. That minimum-depth question is still open.

The extended full-target GRAPE replay frontier does not yet reach the 99% target in the explored 500 ns to 1000 ns window. Every tested seed terminates at the fixed iteration limit, so the present frontier constrains the current implementation settings rather than demonstrating a model-level ceiling.

The strongest SQR winner is also the slowest solution in the final retained set (6180 ns), so its hardware relevance depends sensitively on decoherence times and calibration overhead.

B. Suggested Improvements

- **[P1 — MEDIUM]** Recast the next structured search around the minimum block count that reaches the chosen validation metric at 99%. Within each candidate depth, test orderings first and only then reduce n_{active}.

- **[P1 — HIGH]** Once a depth-minimal structured candidate is identified, perform open-system validation for the retained winners, especially the 6180 ns SQR sequence, and compare their process fidelities directly against the validated GRAPE pulse.

- **[P2 — MEDIUM]** Re-run the strongest SQR and CPSQR structures directly at $N_{\mathrm{cav}} = 12$ from scratch within the selected depth, not only through the staged warm-start path, to check for shorter or even higher-fidelity solutions.

- **[P2 — MEDIUM]** Confirm the local ordering rankings with a second refinement budget or seed set, especially RDS versus DSR for SQR and DRCP versus RDCP/CPDR for CPSQR. The fixed-budget SQR rerun now clearly disfavors DRS, but the ordering sensitivity should still be stress-tested.

- **[P2 — MEDIUM]** After the minimum depth is fixed, search for smaller active-tone sets and then refine duration, leakage, and Wigner agreement. Tone-count reduction should be treated as a follow-up compression step rather than the primary axis.

- **[P2 — MEDIUM]** Add a local waveform-to-state pipeline for the validated GRAPE reference so the appendix can compare retained structured and pulse-level Wigner functions on the same footing.

- **[P2 — MEDIUM]** Test replay-aware or noise-aware GRAPE objectives, preferably with the JAX engine enabled for the current frontier runner. The present GRAPE frontier improves only to replay fidelity 0.9760 at 800 ns and remains iteration-limited across all tested seeds.

- **[P2 — LOW]** Promote the active-subspace diagnostic into shared study utilities or upstream it into `cqed_sim` support code.

C. Open Questions

Why does the SQR family switch from the best screen-median ordering (DSR) to the best fully refined ordering (RDS)? Why does CPSQR ultimately prefer a two-tone DRCP solution even though the strongest coarse-screen replay came from heavier four-tone candidates? Answering these questions may reveal whether the best final structured sequences are exploiting a qualitatively different regime from the one favored by the coarse screen. A second open question now comes from the Wigner analysis: if enlarged spectator-constrained re-fits make both families worse, what physical mechanism is actually responsible for the residual phase-space mismatch?

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[4] N. Ofek, A. Petrenko, R. Heeres, P. Reinhold, Z. Leghtas, B. Vlastakis, Y. Liu, L. Frunzio, S. M. Girvin, L. Jiang, *et al.*, Nature **536**, 441 (2016).

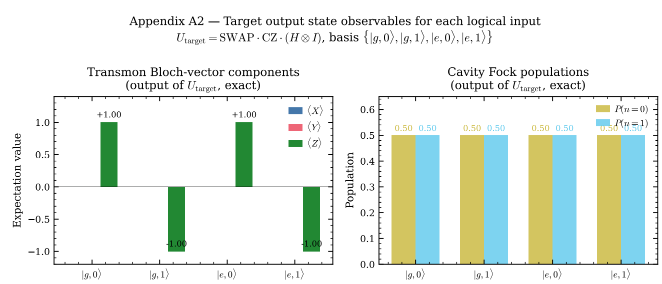

FIG. 6. Exact target output observables for the four logical inputs.

Appendix A: Detailed Results and Data

1. Target observables

Figure 6 shows the exact transmon and cavity observables for the target output states. These are unchanged by the corrected family scope, but they remain the reference signatures for any candidate implementation.

2. Corrected Wigner appendix

Figure 7 now uses a single local figure directory and only shows the target state together with the best retained SQR and CPSQR candidates. No mixed-family panels appear.

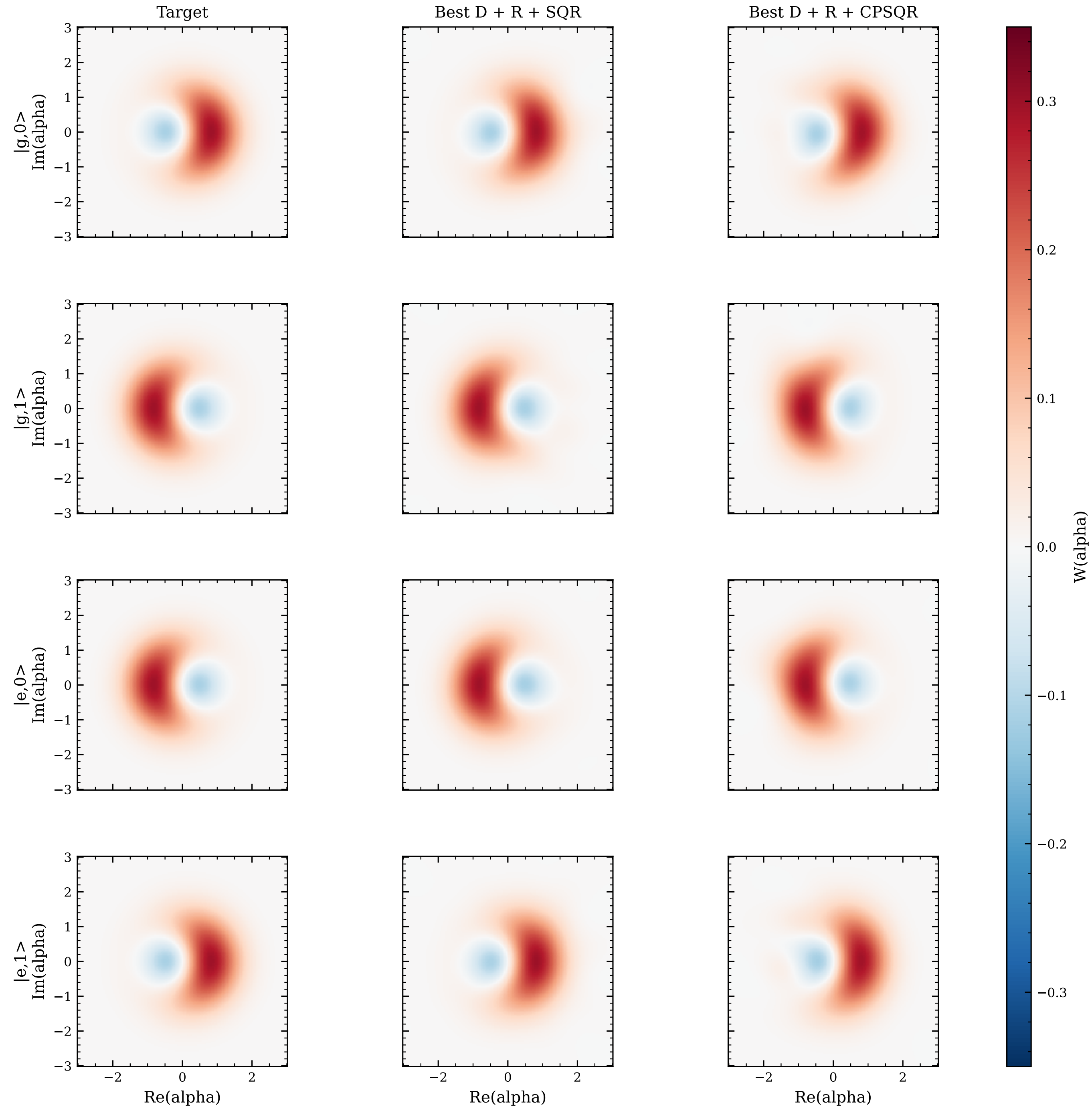

FIG. 7. Appendix Wigner panels from the comprehensive design-space study. The figure shows the target state together with the best retained $D+R+\mathrm{SQR}$ and $D+R+\mathrm{CPSQR}$ candidates.

3. Primitive-level GRAPE diagnostics

To make the primitive-level "third route" concrete within the same study, we optimized one representative retained block from each family with qubit-only GRAPE and compared the replayed action against the corresponding ideal primitive on probe states. The retained SQR representative is the first selective block $S0$ from the final 5-block RDS winner; the retained CPSQR representative is the longest conditional-phase block $CP2$ from the final 5-block $DRCP$ winner. The optimization target is restricted to the diagnostic level subspace relevant to the probe states, while the replay and probe-state comparison are evaluated at $N_{\mathrm{cav}} = 12$.

Figure 8 shows the optimized waveforms and the direct probe-state fidelity comparison. The SQR primitive preserves the reduced cavity state noticeably better than the full joint state, but even its best replay is still far from ideal on the full probe set. The CPSQR primitive is weaker and, most importantly, fails on its addressed $\{1, 2\}$ probe rather than only on a spectator-dominated input. Figure 9 shows the reduced cavity Wigner comparison on the worst reduced-cavity probe for each primitive. These state-level diagnostics make clear that the present primitive GRAPE runs are useful as diagnostics, not yet as drop-in building blocks for the structured cluster-state ansatz.

4. Spectator-constrained Wigner test

The residual Wigner mismatch was probed by re-optimizing each retained family against an augmented target that also constrains spectator sectors. That test does *not* support the hypothesis that the mismatch is caused primarily by unconstrained spectator action. For the retained $D+R+$SQR winner, the deeper logical-only re-fit stays excellent ($F_{12} = 0.999641$, mean Wigner RMS 2.04×10^{-3}), whereas the augmented re-fit degrades to $F_{12} = 0.887459$ with mean Wigner RMS 1.88×10^{-2}. For the retained $D + R +$ CPSQR winner, the deepest complete local run uses a mixed backend path; even there the logical re-fit remains strong ($F_{12} = 0.997589$, mean Wigner RMS 4.70×10^{-3}) while the augmented re-fit falls to $F_{12} = 0.868868$ with mean Wigner RMS 2.82×10^{-2}.

Appendix B: Reproducibility

1. Optimized Parameters

The full gate-by-gate parameters for the retained structured winners are listed here so the paper itself, not only the machine-readable artifacts, records the optimized schedules.

a. *Retained $D + R +$ SQR winner*

This winner uses five selective blocks, four active tones, RDS ordering, and addresses levels $\{0, 1, 2, 3\}$.

R0: QubitRotation, 40.0 ns; $\theta = 0.730$, $\phi = 0.388$; addressed levels: not applicable.
D0: Displacement, 80.0 ns; $\mathrm{Re}(\alpha) = -0.119$, $\mathrm{Im}(\alpha) = 0.302$; addressed levels: not applicable.
S0: MaskedSQR, 1100.0 ns;
$\theta = [-1.879, -1.875, -1.834, 1.486]$,
$\phi = [-1.765, 1.916, 1.689, -1.483]$; addressed levels $0, 1, 2, 3$.
R1: QubitRotation, 40.0 ns; $\theta = 1.486$, $\phi = 1.596$; addressed levels: not applicable.
D1: Displacement, 80.0 ns; $\mathrm{Re}(\alpha) = -0.015$,
$\mathrm{Im}(\alpha) = -0.469$; addressed levels: not applicable.
S1: MaskedSQR, 1100.0 ns;
$\theta = [2.362, -2.107, -2.221, -2.012]$,
$\phi = [-0.427, 0.210, -0.255, -0.506]$; addressed levels $0, 1, 2, 3$.
R2: QubitRotation, 40.0 ns; $\theta = 1.829$, $\phi = 0.237$; addressed levels: not applicable.
D2: Displacement, 80.0 ns; $\mathrm{Re}(\alpha) = 0.285$, $\mathrm{Im}(\alpha) = 0.295$; addressed levels: not applicable.
S2: MaskedSQR, 1100.0 ns;
$\theta = [-0.486, 1.049, -4.545, 3.250]$,
$\phi = [-0.099, 1.143, 1.184, 1.139]$; addressed levels $0, 1, 2, 3$.
R3: QubitRotation, 40.0 ns; $\theta = 1.452$, $\phi = 1.667$; addressed levels: not applicable.
D3: Displacement, 80.0 ns; $\mathrm{Re}(\alpha) = 0.256$, $\mathrm{Im}(\alpha) = 0.137$; addressed levels: not applicable.
S3: MaskedSQR, 1100.0 ns;
$\theta = [0.940, -0.666, 4.213, -3.196]$,
$\phi = [0.059, 0.629, -0.858, -6.539]$; addressed levels $0, 1, 2, 3$.
R4: QubitRotation, 40.0 ns; $\theta = 1.695$, $\phi = -0.135$; addressed levels: not applicable.
D4: Displacement, 80.0 ns; $\mathrm{Re}(\alpha) = 0.336$, $\mathrm{Im}(\alpha) = -0.114$; addressed levels: not applicable.
S4: MaskedSQR, 1100.0 ns;
$\theta = [1.490, -1.220, 19.900, 1.186]$,
$\phi = [-2.205, -2.255, 0.803, 0.879]$; addressed levels $0, 1, 2, 3$.
D5: Displacement, 80.0 ns; $\mathrm{Re}(\alpha) = 0.229$, $\mathrm{Im}(\alpha) = -0.153$; addressed levels: not applicable.

b. *Retained $D + R +$ CPSQR winner*

This winner uses five selective blocks, two active tones, $DRCP$ ordering, and addresses levels $\{1, 2\}$.

D0: Displacement, 80.0 ns; $\mathrm{Re}(\alpha) = 0.273$, $\mathrm{Im}(\alpha) = 0.086$; addressed levels: not applicable.
R0: QubitRotation, 40.0 ns; $\theta = 3.257$, $\phi = -0.062$; addressed levels: not applicable.
CP0: MaskedCPSQR, 111.0 ns; phases = $[1.846, 0.443]$; addressed levels $1, 2$.
D1: Displacement, 80.0 ns; $\mathrm{Re}(\alpha) = -0.304$,
$\mathrm{Im}(\alpha) = -0.409$; addressed levels: not applicable.
R1: QubitRotation, 40.0 ns; $\theta = 2.429$, $\phi = 1.686$; addressed levels: not applicable.
CP1: MaskedCPSQR, 221.0 ns; phases = $[-2.676, -1.085]$; addressed levels $1, 2$.
D2: Displacement, 80.0 ns; $\mathrm{Re}(\alpha) = 0.238$, $\mathrm{Im}(\alpha) = -0.371$; addressed levels: not applicable.
R2: QubitRotation, 40.0 ns; $\theta = 0.433$, $\phi = -1.247$; addressed levels: not applicable.
CP2: MaskedCPSQR, 227.0 ns; phases = $[-1.957, -0.839]$; addressed levels $1, 2$.
D3: Displacement, 80.0 ns; $\mathrm{Re}(\alpha) = 0.527$, $\mathrm{Im}(\alpha) = 0.157$; addressed levels: not applicable.
R3: QubitRotation, 40.0 ns; $\theta = 1.327$, $\phi = 0.900$; addressed levels: not applicable.
CP3: MaskedCPSQR, 154.0 ns; phases = $[-0.012, 2.912]$; addressed levels $1, 2$.
D4: Displacement, 80.0 ns; $\mathrm{Re}(\alpha) = 0.015$, $\mathrm{Im}(\alpha) = -0.007$; addressed levels: not applicable.
R4: QubitRotation, 40.0 ns; $\theta = 1.884$, $\phi = 0.239$; addressed levels: not applicable.
CP4: MaskedCPSQR, 129.0 ns; phases = $[-0.546, -3.532]$; addressed levels $1, 2$.

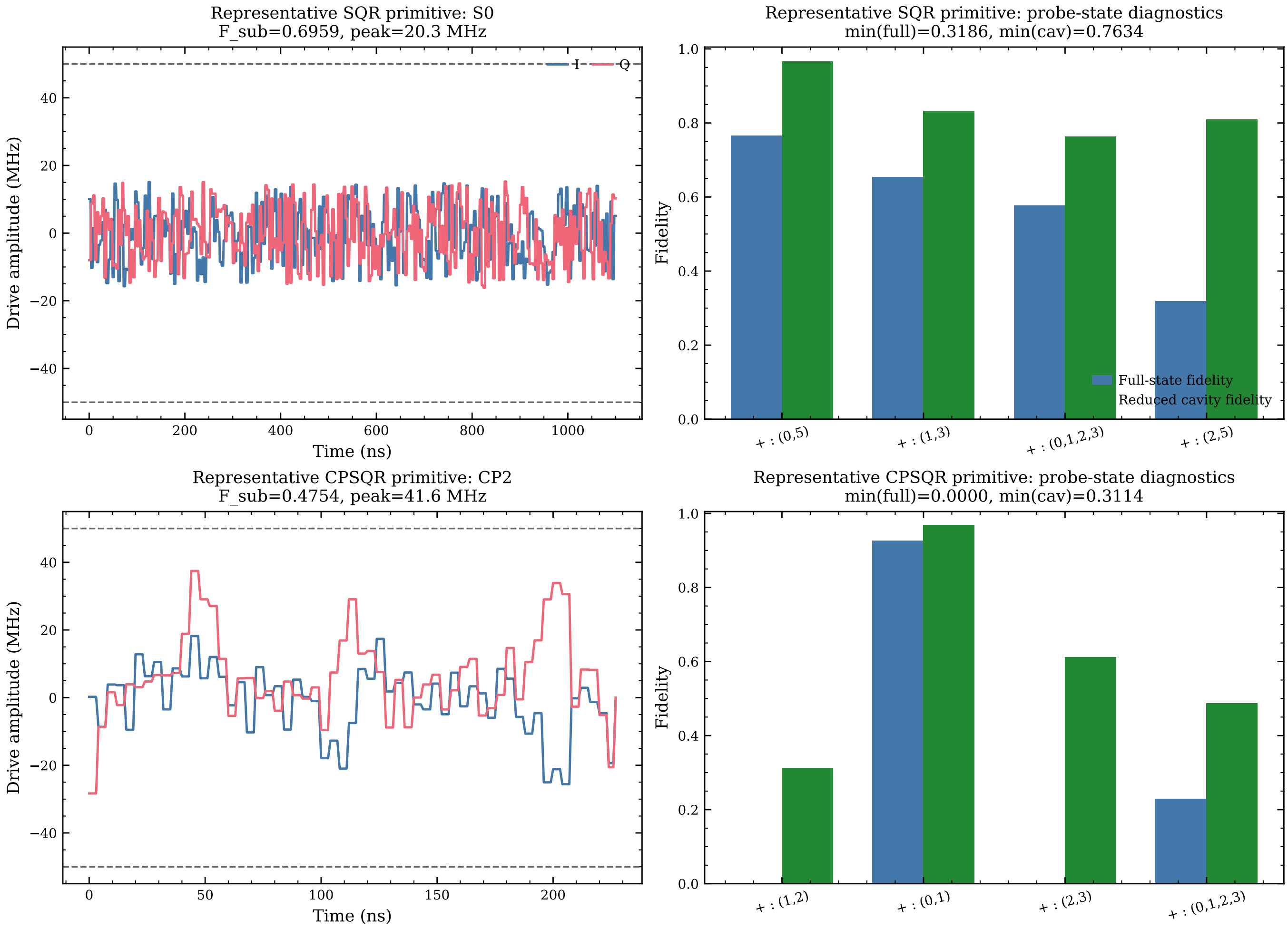

FIG. 8. Representative primitive-level GRAPE diagnostics for the retained exclusive-family winners. Left column: optimized qubit-drive envelopes for the selected $S0$ and $CP2$ primitives, shown in MHz with dashed lines marking the ± 50 MHz amplitude bound. Right column: probe-state fidelities versus the corresponding ideal primitive action. The representative SQR primitive reaches $F_{\mathrm{sub}} = 0.6959$ with mean full-state and reduced-cavity probe fidelities 0.5793 and 0.8429, respectively. The representative CPSQR primitive reaches $F_{\mathrm{sub}} = 0.4754$ with mean full-state and reduced- cavity probe fidelities 0.2892 and 0.5948, respectively.

D5: Displacement, 80.0 ns; $\mathrm{Re}(\alpha) = -0.324$, $\mathrm{Im}(\alpha) = 0.181$; addressed levels: not applicable.

The key machine-readable artifacts are:

- `artifacts/corrected_best_sqr.json`: best preliminary pure SQR candidate after the full design-space refinement; this is also the best overall structured closed-system solution in the study.
- `artifacts/corrected_best_cpsqr.json`: retained pure CPSQR candidate with the full gate parameters, time vector, and physical analysis.
- `data/grape_frontier_extension.json`: validated full-target GRAPE frontier for 500 ns, 600 ns, 800 ns and 1000 ns, including per-seed replay metrics and open-process fidelities.
- `data/primitive_grape_diagnostics.json`: representative primitive-level GRAPE diagnostics for the retained $S0$ and $CP2$ blocks, including per-seed subspace fidelities and probe-state metrics.
- `artifacts/primitive_grape_diagnostics_arrays.npz`: saved waveform and Wigner arrays backing the primitive diagnostic figures.
- `data/corrected_scope_summary.json`: complete design-space search, finalist refinement, physical retention, active-subspace diagnostics, and GRAPE reference.
- `data/corrected_design_space_best.csv`: one row per structural configuration from the 216-point screen, with replay metrics at $N_{\mathrm{cav}} = 12$.
- `data/corrected_target_99_finalists.csv`: final

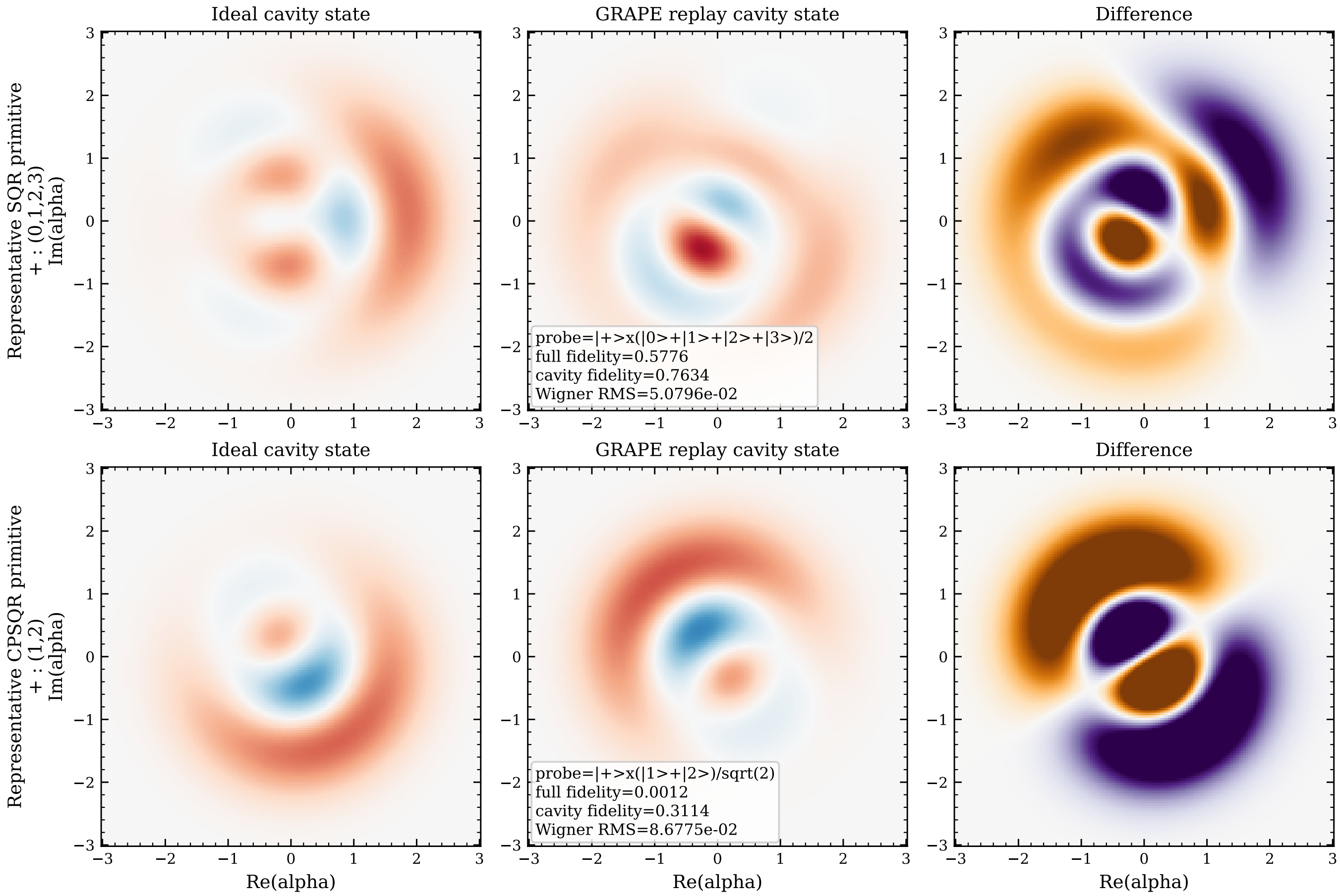

FIG. 9. Reduced-cavity Wigner comparison on the worst reduced-cavity probe for each representative primitive. Top: the retained SQR primitive $S0$ evaluated on $|+\rangle \otimes (|0\rangle + |1\rangle + |2\rangle + |3\rangle)/2$, where the replayed primitive reaches full-state fidelity 0.5776 and reduced-cavity fidelity 0.7634. Bottom: the retained CPSQR primitive $CP2$ evaluated on $|+\rangle \otimes (|1\rangle + |2\rangle)/\sqrt{2}$, where the replayed primitive falls to full-state fidelity 0.0012 and reduced-cavity fidelity 0.3114. The phase-space mismatch is therefore moderate for SQR and severe for CPSQR.

retained, low-confidence, and discarded physical finalists after the $N_{\mathrm{cav}} = 12$ re-optimization.

- `data/what_changed_summary.md`: short revision summary versus the previous unified report.

2. Modeling and Simulation Assumptions

The structured families are screened at $N_{\mathrm{cav}} = 4$ and physically re-optimized at $N_{\mathrm{cav}} = 12$ for a shortlisted finalist set, then validated at $N_{\mathrm{cav}} = 10, 12, 14$ with $N_{\mathrm{tr}} = 2$ transmon levels. The final workflow consists of 216 structural screen cases, 30 level-subset refinements, and 12 physical finalists. The active-subspace rule uses 12 fractional samples per gate and a population threshold 10^{-3} with a 99.9% capture requirement.

3. Reproduction Procedure

From the study `scripts/` directory:

1. Run `python run_design_space_study.py` to regenerate the design-space summary, best-candidate artifacts, and all local figures.
2. For figure-only regeneration, run `python run_design_space_study.py --skip-search`.
3. Compile the report from `report/` with `pdflatex report.tex; bibtex report; pdflatex report.tex; pdflatex report.tex`.

Agreement checks:

- The best pure SQR candidate should report $F_{12} \approx 0.99963$, worst leakage $\approx 7.8 \times 10^{-4}$, and active support bounded to levels 0 through 3.
- The best pure CPSQR candidate should report $F_{12} \approx 0.99749$, worst leakage $\approx 5.4 \times 10^{-3}$, and active support bounded to levels 0 through 5.

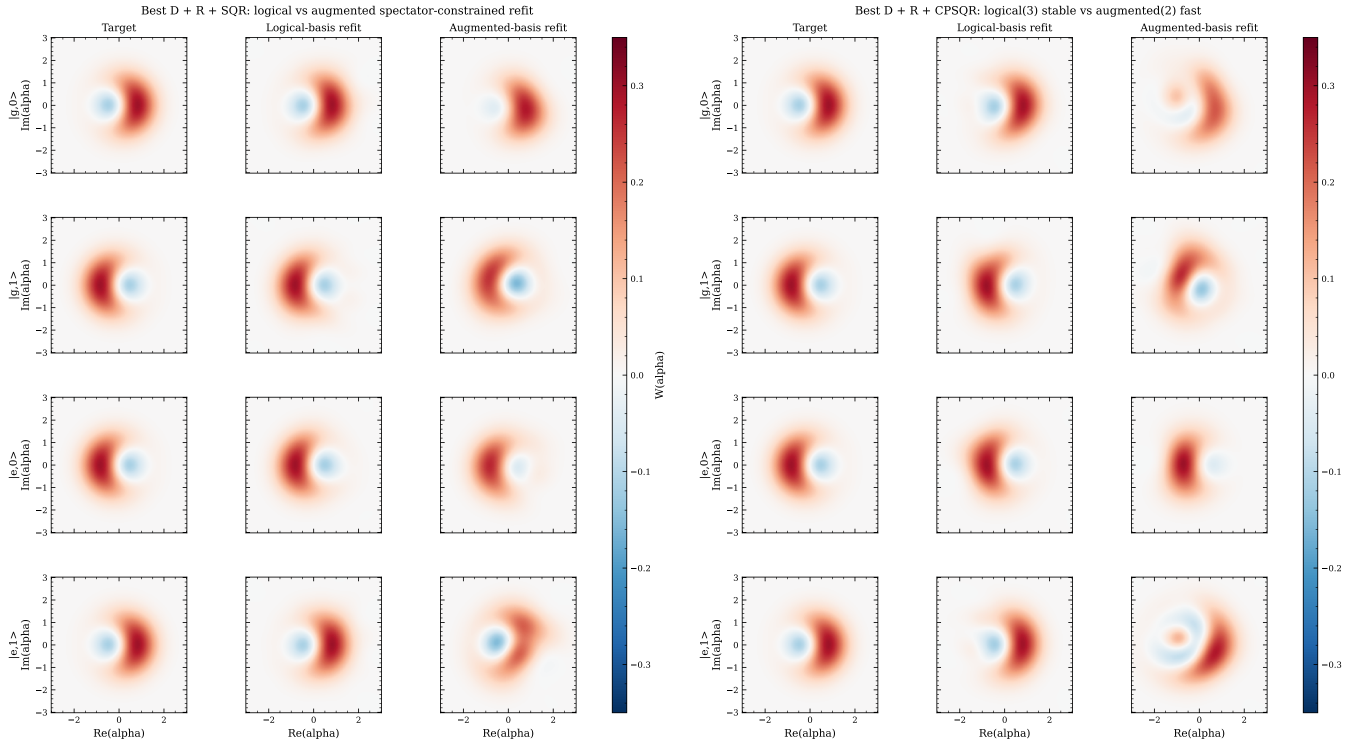

FIG. 10. Deeper spectator-constrained Wigner tests for the retained exclusive families. Left: the retained $D+R+\mathrm{SQR}$ winner. Right: the deepest complete local $D+R+\mathrm{CPSQR}$ test, which uses a mixed backend path for numerical stability. In both families the spectator-constrained augmented target worsens both fidelity and phase-space agreement rather than improving them.